

Activity

Group members:

Grap applications data

Suppose we are interesting in testing whether there is a relationship between college GPA and acceptance to a graduate program. We fit the regression model

$$Admit_i \sim Bernoulli(\pi_i)$$

$$\log\left(\frac{\pi_i}{1-\pi_i}\right) = \beta_0 + \beta_1 GPA_i$$

and we wish to test the hypotheses $H_0 : \beta_1 = 0$ vs. $H_A : \beta_1 \neq 0$. We will use the drop-in-deviance statistic

$$G = \text{deviance}_{\text{reduced}} - \text{deviance}_{\text{full}}$$

Here is model output for the full and reduced models, and a histogram approximating the null distribution for the drop-in-deviance statistic G :

```
grad_app <- read.csv("https://stats.idre.ucla.edu/stat/data/binary.csv")

# full model
glm(admit ~ gpa, family = binomial, data = grad_app)
```

Call: glm(formula = admit ~ gpa, family = binomial, data = grad_app)

Coefficients:

(Intercept)	gpa
-4.358	1.051

Degrees of Freedom: 399 Total (i.e. Null); 398 Residual

Null Deviance: 500

Residual Deviance: 487 AIC: 491

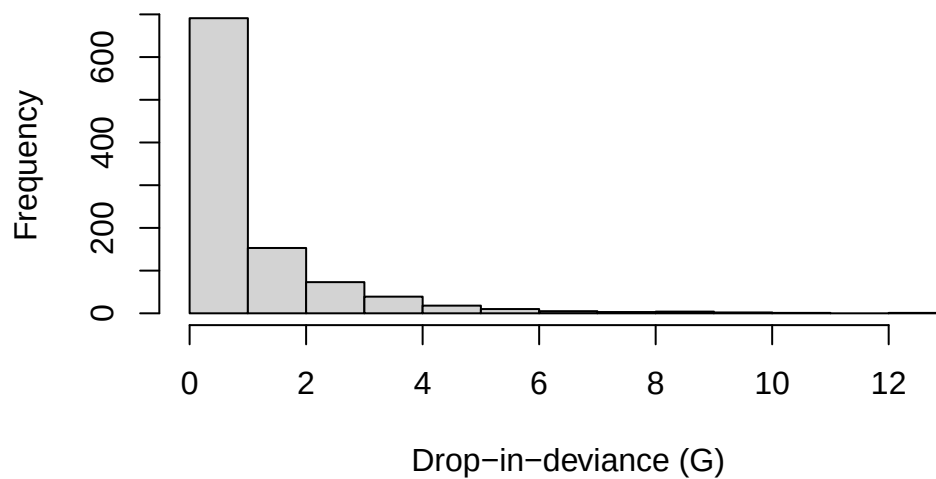
```
# reduced model  
glm(admit ~ 1, family = binomial, data = grad_app)
```

Call: `glm(formula = admit ~ 1, family = binomial, data = grad_app)`

Coefficients:
(Intercept)
-0.7653

Degrees of Freedom: 399 Total (i.e. Null); 399 Residual
Null Deviance: 500
Residual Deviance: 500 AIC: 502

Approximate null distribution



1. What do you notice about the “Null Deviance” for the two models? How does it compare to the “Residual Deviance”?
2. Calculate the drop-in-deviance statistic for the hypotheses above.
3. Based on the observed drop-in-deviance and the null distribution, do you think that the null hypothesis is true?